HW IV

EE 588: Optimization for the information and data sciences

University of Southern California

Assigned on: October 13, 2026 Due date: beginning of class on October 28, 2026

The following exercises from the text: 6.2, 6.10, 7.1 and the following additional exercises

- *Total variation image interpolation.* A grayscale image is represented as an $m \times n$ matrix of intensities U^{orig}. You are given the values U_{ij}^{orig}, for $(i,j) \in \mathcal{K}$, where $\mathcal{K} \subset \{1,\ldots,m\} \times \{1,\ldots,n\}$. Your job is to *interpolate* the image, by guessing the missing values. The reconstructed image will be represented by $U \in \mathbb{R}^{m \times n}$, where U satisfies the interpolation conditions $U_{ij} = U_{ij}^{\text{orig}}$ for $(i,j) \in \mathcal{K}$.

  The reconstruction is found by minimizing a roughness measure subject to the interpolation conditions. One common roughness measure is the ℓ_2 variation (squared),

  $$\sum_{i=2}^{m} \sum_{j=2}^{n} \left((U_{ij} - U_{i-1,j})^2 + (U_{ij} - U_{i,j-1})^2 \right).$$

  Another method minimizes instead the *total variation*,

  $$\sum_{i=2}^{m} \sum_{j=2}^{n} \left(|U_{ij} - U_{i-1,j}| + |U_{ij} - U_{i,j-1}| \right).$$

  Evidently both methods lead to convex optimization problems.

  Carry out ℓ_2 and total variation interpolation on the problem instance with data given in `tv_img_interp.m`. This will define `m`, `n`, and matrices `Uorig` and `Known`. The matrix `Known` is $m \times n$, with (i,j) entry one if $(i,j) \in \mathcal{K}$, and zero otherwise. The mfile also has skeleton plotting code. (We give you the entire original image so you can compare your reconstruction to the original; obviously your solution cannot access U_{ij}^{orig} for $(i,j) \notin \mathcal{K}$.)

- *Fitting with censored data.* In some experiments there are two kinds of measurements or data available: The usual ones, in which you get a number (say), and *censored data*, in which you don't get the specific number, but are told something about it, such as a lower bound. A classic example is a study of lifetimes of a set of subjects (say, laboratory mice). For those who have died by the end of data collection, we get the lifetime. For those who have not died by the end of data collection, we do not have the lifetime, but we do have a lower bound, i.e., the length of the study. These are the censored data values.

  We wish to fit a set of data points,

  $$(x^{(1)}, y^{(1)}), \ldots, (x^{(K)}, y^{(K)}),$$

with $x^{(k)} \in \mathbb{R}^n$ and $y^{(k)} \in \mathbb{R}$, with a linear model of the form $y \approx c^T x$. The vector $c \in \mathbb{R}^n$ is the model parameter, which we want to choose. We will use a least-squares criterion, i.e., choose c to minimize

$$J = \sum_{k=1}^{K} \left(y^{(k)} - c^T x^{(k)} \right)^2 .$$

Here is the tricky part: some of the values of $y^{(k)}$ are censored; for these entries, we have only a (given) lower bound. We will re-order the data so that $y^{(1)}, \ldots, y^{(M)}$ are given (i.e., uncensored), while $y^{(M+1)}, \ldots, y^{(K)}$ are all censored, i.e., unknown, but larger than D, a given number. All the values of $x^{(k)}$ are known.

1. Explain how to find c (the model parameter) and $y^{(M+1)}, \ldots, y^{(K)}$ (the censored data values) that minimize J.
2. Carry out the method of part (a) on the data values in `cens_fit_data.m`. Report $\hat{c}$, the value of c found using this method.

   Also find $\hat{c}_{\mathrm{ls}}$, the least-squares estimate of c obtained by simply ignoring the censored data samples, i.e., the least-squares estimate based on the data

   $$(x^{(1)}, y^{(1)}), \ldots, (x^{(M)}, y^{(M)}).$$

   The data file contains c_{true}, the true value of c, in the vector `c_true`. Use this to give the two relative errors

   $$\frac{\|c_{\mathrm{true}} - \hat{c}\|_2}{\|c_{\mathrm{true}}\|_2}, \qquad \frac{\|c_{\mathrm{true}} - \hat{c}_{\mathrm{ls}}\|_2}{\|c_{\mathrm{true}}\|_2}.$$